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Source: *The Quarterly Journal of Economics*, Aug., 1990, Vol. 105, No. 3 (Aug., 1990), pp. 745-771

Published by: Oxford University Press

Stable URL: <https://www.jstor.org/stable/2937897>

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THE ALLOCATION OF RESOURCES BY VOTING*

YORAM BARZEL AND TIM R. SASS

A general theory of voting, which explains under what conditions voting will be chosen as a means for allocating resources and how the constitution that governs the voting will be structured is presented. It is hypothesized that developers of voting organizations will structure their organizations in order to maximize the value of shares sold by minimizing the expected costs of wealth transfer and decision making in the voting organization. Implications regarding the allocation of votes and assessments within the organization, the domain of voting decisions, and the optimal voting rule are tested with data on the constitutional structure of condominium homeowner associations.

Voting is used by many organizations, both public and private, to allocate scarce resources. Economic analysis of the structure of such voting organizations has developed in two essentially independent bodies of literature. The "public choice" literature, based largely on the seminal work of Buchanan and Tullock [1962], focuses on the structure of political voting organizations. Assuming individual maximization, Buchanan and Tullock analyze the choice between voluntary action in the market and political action. Their theory of constitutional structure focuses on the choice of the optimal voting rule as a function of the characteristics of the decision-making group, and of the type of decision to be made. In the twenty-five years since the initial work of Buchanan and Tullock was published, only relatively minor critiques and extensions of their theory have been advanced in the public choice literature.¹

In contrast, the corporate voting literature, initiated early by Manne [1964], has recently blossomed with the work of Easterbrook and Fischel [1983], Nitzan and Procaccia [1986], and particularly Grossman and Hart [1988] and Harris and Raviv [1988]. This work has focused primarily on two issues: the assignment of votes in relation to income claims within a corporation, and the selection of

*We wish to thank Robert Ellickson and Barry Weingast for their valuable comments. We also received helpful comments from Doug Allen, Linda DeAngelo, Frank Easterbrook, Micha Gisser, Tom Goodwin, John Hause, Michael Pogodzinski, Raymond Sauer, Pablo Spiller, and an anonymous referee. Thanks also go to Keith Burris for excellent research assistance, and the Research Allocation Committee of the University of New Mexico for providing financial support.

1. For example, see Kiesling [1968], Östrom [1968], and Kafoglis and Cebula [1981].

the voting percentage required to transfer control of a corporation. The most recent work in this area, that of Grossman and Hart and of Harris and Raviv, has concentrated on the effect of private benefits accruing to management from corporate control on the optimal choice of these two variables.

A general theory of voting should explain under what conditions voting will be chosen as a means for allocating resources, and how the constitution that governs the voting, including the allocation of votes and assessments within the organization, will be structured. The existing public choice literature, while applicable to a wide range of voting organizations with varying collective choice problems, has concentrated largely on the voting rule, thereby neglecting to effectively address of the many other aspects of a voting institution. In contrast, the relatively new corporate voting literature does consider both the voting rule and the assignment of votes, but it does so only in the narrow context of corporate control decisions. In this paper we explain the choice of the various components of a voting organization's constitution in a general context that is applicable to both political and business voting.

In addition to developing a general economic theory of constitutional choice, we provide a number of tests of our theory. No such empirical analysis of the initial choice of rules under which voting is to take place has been previously conducted.² Two factors seem responsible for this lack of empirical work on constitutional choice: first, the difficulty in untangling the initial maximization process in political constitutions; and second, the relative homogeneity of vote assignments within the types of voting organizations previously studied.

Although all voting institutions are erected by the same maximizing individuals who operate in markets, and are therefore endogenous, untangling the endogeneity of voting mechanisms in large political units is problematic. Tracing the political entity's historical origins back to the introduction of its constitution—often an impossible task—is required. The study of this seemingly intractable political problem, however, becomes relatively simple

2. Political scientists have conducted some empirical studies of the adoption of "reform" elements in the charters of municipal governments. These elements include electing council members at large, holding nonpartisan elections, and employing city managers rather than elected mayors. However, such studies use indices of the reform elements as the dependent variable in their analyses, rather than specific individual provisions. See Farnham and Bryant [1985] and Maser [1985].

when one observes a business venture that incorporates voting as part of its decision-making mechanism. The criteria for maximization are much easier to delineate in such a private voting organization. Given wealth maximization by its developer, he will select the stipulations of the organizations's constitution that are expected to maximize the net present value of the venture to its prospective buyers.³

A second factor contributing to the dearth of empirical analysis is the fact that both purely political bodies and corporations, taken separately, have relatively little diversity in the assignment of voting rights. In political organizations, the one-man-one-vote rule is used almost exclusively, while one-share-one-vote dominates in the corporate voting context.⁴ This makes analysis of the choice of voting rights within either type of organization difficult. Fortunately, this problem can also be overcome by studying private, noncorporate organizations that use the voting mechanism for decision-making purposes.

In order to avoid the problems mentioned above, we have chosen to focus our analysis on owners' associations of newly developed condominiums. Such private "governments" are particularly well suited to the study of voting organizations for a number of reasons. First, the constitution of the association is written by the condominium's developer, making the maximand clear. Second, as the association emerges immediately in its "fully grown" state, its evolution is easy to trace. Third, the assignment of voting rights among condominium owner associations is relatively diverse. Fourth, in contrast to purely political voting where benefits are largely nonpecuniary, and corporate voting where benefits are almost exclusively pecuniary, benefits in condominium associations include both pecuniary and nonpecuniary elements, thereby allowing an analysis of the form of benefits on institutional structure. Fifth, and perhaps most important, the study of such associations is

3. The idea that the constitution is part of the maximization process is not new. Grossman and Hart [1980, p. 48], for instance, say "... at the time the [corporation's] charter is written, initial shareholders try to devise self-enforcing mechanisms to ensure good management."

4. Out of the thousands of corporations with publicly traded stock in 1980, DeAngelo and DeAngelo [1985] could identify only 78 corporations that assigned votes on other than a one-share-one-vote basis. See also Lease, McConnell, and Mikkelsen [1983], who measure the relative values of voting rights in such dual class stock corporations, and Partch [1987] and Jarrell and Poulsen [1988] who estimate the effect on the market value of stock when a second class of stock with differing voting rights is issued.

empirically tractable since their "constitutions" and relevant physical properties are a matter of public record.

In Section I we begin by analyzing the choice between market organization and the use of voting. It is shown that the use of voting to allocate resources can avert certain moral hazard opportunities associated with market transactions. Voting entails two kinds of costs: those of decision making and those associated with wealth transfer. These voting costs are the subject of Section II. In Section III we analyze the ways in which the developer of a voting organization can structure its constitution in order to minimize the voting costs related to wealth transfers, and thereby maximize his gains. Finally, in Section IV many of the predictions formulated in previous sections, as well as some implications of the Buchanan-Tullock model, are tested using data on condominium homeowner associations.

I. A RATIONALE FOR VOTING ORGANIZATIONS

In many organizations that allocate resources by voting, voters are both the owners of resources used by the organization and the consumers of its services. By internalizing exchange, such voting organizations avoid certain transaction costs that would be incurred with market exchange. There is a trade-off, however, since operating a voting mechanism also consumes resources.

To illustrate the cost of market exchange, consider a project requiring a large, centrally used amount of capital. When the requisite capital exceeds an individual's personal wealth, he may go to the market and borrow others' capital, paying fixed interest on the loan and becoming the sole residual claimant. It is well-known that forming and policing loan contracts is costly. By undertaking investment projects that are subject to a lower expected return and higher variability than those of other available projects, the borrower can gain at his lenders' expense. The precautions lenders will take in order to reduce their expected losses are also costly. On the margin, lenders will spend one dollar in resources to avert the transfer of one dollar of their wealth to borrowers. Borrowing, a market exchange, therefore involves policing and other transaction costs.⁵

5. The costs of using the market include the costs of writing and enforcing contracts involving contingent claims. While such contracts may prevent some wealth transfers, they must be written *ex ante*, when it is costly to determine all

Alternatively, the suppliers of capital can invest in equity. As long as all equity holders have the same rights and are subjected to the same costs, there will not be a divergence in incentives, as there is with debt financing. Since the suppliers of capital also decide how it will be employed, no exploitation within the group will take place. However, equity holders require a decision-making mechanism for the collective use of their capital, which allows each of them to remain an owner. Voting, appropriately designed, is a method by which to achieve simultaneously joint ownership and decision making. The use of voting thereby replaces price (interest rate here) as the mechanism for allocating resources.

We can view the choice of voting to allocate resources in another way. Implementing detailed pricing and policing is costly. Market exchange will thus leave some valued commodity attributes in the public domain. Whereas resources spent to effect and to avert transfers are "dissipated," the dissipation can be reduced by "harmonizing" the interests of the two parties.⁶

The problems that voting organizations may eliminate are general; they are present in any exchange. In order to make the model operational, conditions under which voting organizations rather than markets will be used must be specified, either by delineating the characteristics of exchanges that are relatively more prone than others to costly attempts to capture wealth, or by determining circumstances under which the voting organization's costs decline while the gains they generate remain constant. As economists, we lack the expertise to delineate characteristics of commodities that are relatively costly to exchange in the market. We are better equipped to identify factors that affect the costs of voting.

II. THE COSTS OF VOTING

Maximization of the net gains from a voting organization implies structuring the organization in order to minimize the two types of costs associated with voting. The first type includes the resources expended in attempts to use the voting process to transfer

possible contingencies. Easterbrook and Fischel [1983] also view voting as an alternative to costly contingent contracts. They say, "voting exists in corporations because someone must have the residual power to act (or delegate) when contracts are not complete" [Easterbrook and Fischel, 1983, p. 403].

6. The dissipation considered here is of the same logical nature as the dissipation from excess measuring. See Barzel [1982].

wealth among members in the organization. In their discussion of this wealth-transfer problem, Buchanan and Tullock focus on the losses that an *individual* expects to incur as a result of others' actions in the voting process, in particular being subjected to voting decisions that go against his interest when in the minority. Many of these costs tend to cancel out across individuals, however; a loss suffered by someone in the minority may be offset by a gain to a member of the majority.⁷ If losses due to potential wealth transfers were only of this distributional nature, they would not affect the decision making of a developer seeking to maximize the *total* net value of the voting organization.

Of greater concern to a maximizing developer is the value of resources that voters expect to expend in attempts to effect and avoid wealth transfers among themselves. In order either to transfer one dollar of wealth to themselves or to avoid the transfer of one dollar of their wealth to others, voters will be willing to spend up to one dollar worth of resources. Since voters will pay a developer more when they expect to devote fewer resources to wealth capture and prevention, a developer will attempt to reduce the potential for redistribution through voting.

The second type of voting cost, called "decision-making costs" by Buchanan and Tullock, entails both the effort devoted to reaching decisions by voting, as well as the costs of strategic bargaining by voting members.⁸ Strategic bargaining costs are the losses imposed by individual members who prevent agreement by holding out for a disproportionate share of the gains.

Yet another component of decision-making costs (unaccounted for by Buchanan and Tullock) is the cost of acquiring information required to make voting decisions. An individual will collect only that information which he values more than his cost of acquiring it. The information any single voter requires to guide his voting decision is similar to that needed by all other voters; individual voters will not collect information that can be obtained more cheaply by copying fellow voters.⁹ This consideration applies to all

7. The losses from wealth transfers, termed "external costs" by Buchanan and Tullock, also include deadweight costs due to inefficient levels of public activities selected by the majority as well as pure transfers. In addition, risk-averse individuals may prefer not to be subject to the uncertainty of possible transfers. We do not attempt to capture either of these effects in our analysis.

8. These two components of decision-making costs, not explicitly delineated by Buchanan and Tullock, are emphasized by Kiesling [1968].

9. This point is made in the corporate voting context by Manne [1964].

voters, and thus some useful information may not be collected at all, while the effort to acquire other information may be duplicated.

Voting organizations are expected to seek methods that minimize the voter information problem. For instance, publicly held corporations furnish their stockholders with information by providing financial and auditor's reports, and data pertaining to directors. Corporations also reduce the information problem by hiring managers who are, in part, information specialists. Such information specialists, by the nature of their function, possess more information than their employers, however, and may attempt to exploit their advantage. Therefore, voters must still collect some information on their own in order to be able to assess the performance of specialists, and decide how to vote.

Voters who do acquire information make more informed choices, the gains from which are shared by all group members. In a shareholder firm, stockholders should therefore be rewarded not only for furnishing capital, but also for contributing to decision making. The incentive to collect information and vote increases with the size of the individual's stock ownership, since the costs are independent of that ownership and the return is proportional to a person's share holdings. It is expected, then, that large holders will both collect more information and be compensated for their extra contribution.¹⁰

III. THE STRUCTURE AND CONSTITUTION OF VOTING ORGANIZATIONS

Three factors determine a buyer's valuation of an ownership share in a jointly held asset controlled by voting: the value of the asset at the time of purchase (how attractive is the condominium's swimming pool?), his perception of the subsequent time path of changes in the asset's value (will the pool be well maintained?), and

10. Consistent with these expectations, Brickley, Lease, and Smith [1988] find that the percentage of stock voted in antitakeover votes is positively related to the proportion of stock held by nonmanagement owners of large blocks of stock. Nitzan and Procaccia [1986] reason that the one-share-one-vote rule provides such large shareholders a proper incentive to acquire information about voting issues. Also, see Easterbrook and Fischel [1983] who recognize that the right to vote is associated with stock return.

The refund policies of some cooperatives also seem to reward voters' effort. They provide their services to nonmembers and to members on equal terms, yet members (voters) receive "refunds" while nonmembers get smaller refunds or none at all [Bogen, 1968, 12.11], thereby enhancing members' incentives to contribute to the cooperative management.

his expected voting costs associated with the asset (how costly will a dispute over guests' pool use be?). Buyers' valuations of the assets offered by a developer therefore depend, in part, on their expectation of how deciding majorities of members will vote in future periods, and on their own expected expenses in attempts to affect those voting decisions.

If voters' costs and gains were identical, a decision favored by one would be favored by all: votes would be unanimous, and exploitative opportunities absent.¹¹ In such cases buyers would offer higher amounts than they would offer in instances where votes are contested and wealth transfers occur. Such increases in valuation can be extracted from buyers by developers in the form of higher prices. Developers, then, will gain by structuring their offerings so that buyers' interests are in harmony. We first consider the requisite conditions for harmonizing buyers' interests,¹² and then analyze the methods by which developers can achieve such harmony.

A. Harmonizing Voters' Interests and the Form of Organizational Benefits

If individuals' shares yield uniform net gains, their interests would be in harmony. Uniform net gains and the corresponding harmony of interests are obviously achieved when similar individuals are provided with identical facilities and are assessed equally. However, harmony can also be achieved under less stringent conditions; it suffices that voters agree on issues that come up for vote, even if they disagree on other matters. Harmony is relatively easy to attain in projects that provide only pecuniary benefits, but harder to achieve when projects also provide their owners direct consumption.

Assets are valued for both directly contributing to consumption and for the pecuniary gains they may generate. The larger the proportion of the pecuniary gains in the total valuation of an asset, the easier it is to reach agreement concerning the asset's performance, since then valuation depends on objective criteria. When all returns from an asset are pecuniary and known with certainty, shares that yield uniform net gains can be readily formed, and

11. Decision-making costs, in particular the costs associated with strategic bargaining, would still be present, however.

12. It may appear that the simplest method of attaining harmony of interests is by letting buyers select themselves. Self-selection will obviously attract individuals who value the asset highly. Among these, however, are adverse selectors—individuals who value the asset highly because they can gain at others' expense.

owners are likely to agree on their valuation of the asset. Developers of public shareholder firms are able to achieve such agreement among shareholders most successfully.

Corporate shares possess the following harmony-inducing features: (1) rewards to shareholders are uniform across shares, since shares are valued for their claims to dividends and changes in their market price, rather than for any direct contribution to utility; (2) the buyers of corporate shares are allocated votes in proportion to their share holdings;¹³ and (3) assessments are zero. Therefore, any project demonstrably enhancing the value of the corporation will receive unanimous support.¹⁴ The developers of corporations are able to structure their assets and the rights of prospective owners such that there are almost no opportunities for majorities to capture wealth from minorities.¹⁵ In fact, in most stockholder votes the margin of victory is overwhelming.¹⁶ Correspondingly, prospective owners' valuations of the shares offered by a developer of a stock corporation are not diluted by expectations of future resource expenditures to effect wealth transfers among themselves.

Buyers into projects that generate consumption services are also the final consumers of the services and thus obtain utility directly from their purchases.¹⁷ The provision of commodities that contribute to individuals' utilities is difficult to harmonize, as such commodities are in general harder to evaluate and more costly to trade than projects contributing to pecuniary wealth. It is therefore more difficult for developers of projects producing consumption

13. Both Harris and Raviv [1988] and Grossman and Hart [1988] demonstrate that when private benefits from corporate control, such as perquisites, are small, the selection of uniform income claims and voting rights (i.e., one share, one vote) and a majority voting rule maximizes the total value of the firm's securities. Further, Grossman and Hart show that only when there exist significant private benefits from corporate control for both present management and potential management are deviations from one-share-one-vote optimal. This is consistent with our theory in that votes not based on income claims are desirable only when benefits are not strictly proportional to income claims.

14. DeAngelo [1981] demonstrates conditions for unanimity. Three types of decisions came to mind regarding which unanimity may not be achieved in reality: the distribution of dividends, dealings of the corporation with other firms owned by subsets of the stockholders, and decisions regarding management when managers are shareholders.

15. The primary exception is where majority control of a corporation can allow management to extract private benefits at the expense of other shareholders. See Grossman and Hart [1988, pp. 191-95].

16. Shareholders' ignorance or indifference is likely to reduce the participation rate, but seems to be unrelated to the degree of unanimity.

17. Partnerships such as lawyers or physicians are intermediate organizations in that although they produce services for the general public, the effect of working conditions on partners' utility seems to be of particular importance. Fama and Jensen [1983a, 1983b] analyze the structure of such professional partnerships.

services than for developers of stock firms to secure harmony of interest.

Such is the case in political voting where benefits received by voters in a political jurisdiction are primarily nonpecuniary, coming in the form of goods and services provided by government. Because such benefits are mostly nonpecuniary, harmonizing voter interests would be difficult even if that were the policy objective.¹⁸ With heterogeneous preferences, voters' interests may diverge, creating opportunities for majorities to capture wealth from minorities through majority voting.

The more diverse the interests of voters, the greater the potential for contested voting decisions, and for the corresponding costs of potential wealth transfers. For this reason, we expect that in order to protect prospective buyers from wealth capture, developers of utility-generating projects will choose more elaborate constitutional provisions than those made by developers of cash-generating projects. For example, the interests of owners who purchase condominium units as investments generating tax advantages rather than as residences are for the most part confined to the market value of their units. The constitutions of such condominiums are therefore expected to be less restrictive, including fewer precautions against transfers, than the constitutions of owner-resident condominiums.¹⁹ Our theory similarly predicts that the constitutions of commercial condominiums will be less restrictive than those of residential condominiums, since the gains to unit owners in commercial condominiums are primarily pecuniary. In particular, our theory predicts that the voting rule selected by developers of commercial condominiums will be less inclusive, i.e., farther from unanimity, than that chosen by developers of residential condominiums. This implication will be tested in Section IV below.

Maximizing the present value of shares in a voting organization requires that harmony be maintained over time. The ability to maintain harmony will vary, as before, between assets generating pecuniary gains and those providing primary consumption gains. For example, the public stock corporation's objective of wealth maximization remains unaltered with the passage of time. Even if it

18. In the same spirit Manne [1964, p. 1441] states that "a crucial difference between political and corporate elections [is that] with very rare exceptions, there are no conflicts in the corporate interests of voting shareholders."

19. The more closely owners represent their tenants' interests, the higher the rent they can collect. However, since owners are not expected to be perfectly well informed of their tenants' preferences, the prediction in the text is still valid.

is expected that opinions on future issues will be divided, the low-cost transferability of shares makes it likely that current owners will be in agreement on the issues at hand at any given time.²⁰ Harmony is more difficult to sustain in voting organizations providing nonpecuniary benefits. Take the example of homeowners' associations. Neither homes nor owners age at a uniform rate. A change in ownership can avert exploitation based on differences in personal characteristics; the tendency of families with children to push for the construction of a playground to be funded in part by assessments on childless homeowners is constrained by the latter's ability to sell their units to families with children. While ownership turnover is expected to enhance agreement, the complexity of a dwelling makes it costly to sell; disagreement among owners will therefore continue. Anticipating significant future conflicts, the initial valuation of the units by prospective buyers will be correspondingly lower. In order to raise units' values, we expect the developer to enhance the role of supermajorities and other restraints on wealth capture that might occur as the organization ages. We also predict that a low-cost mechanism to dissolve the organization, or to withdraw from it, will be included in the organization's constitution.²¹

B. Developers' Maximizing Actions

A developer has three major organizational tools with which to harmonize interests and thereby maximize the net present value of his project. The first is the choice of ownership rights that buyers receive with the asset. In particular, the assignment of assessments and voting rights will have a direct impact on the harmony of interests, and hence on the costs of voting. The second is the developer's control of the features of the transferred asset; he can decide ahead of time which features will be outside the domain of subsequent voting and which will be left to the discretion of the voting organization. The third is the rules under which decisions are made. Voting rules that are more inclusive (i.e., nearer unanimity) lower the potential for wealth transfers at the expense of higher decision-making costs.

In addition to its obvious effect on buyers' valuation of units,

20. This point was first made by Manne [1964]. The option of sale decreases the role of voting in the decision-making process. See Easterbrook and Fischel [1983].

21. Fama and Jensen [1983a, 1983b] emphasize the use of redeemable claims in financial mutuals, where shareholders can withdraw their assets on demand, as an important mechanism for controlling agency problems in these institutions.

the assignment of assessments will also affect the degree to which buyers will agree on voting issues. The more voters agree, the fewer resources that will be expended in attempts to effect and prevent wealth transfers. A necessary condition for achieving unanimity among voters is that a project generating net positive value to one generates positive value to all.²² When the total net gain from a project is positive, there exists at least one set of assessments that yields a net positive value to each participant. However, implementing an assessment scheme possessing such properties encounters some problems. First, if potential projects differ in the distribution of benefits they generate for owners, each project would require its own assessment formulae in order to achieve unanimity on all voting issues. Construction and implementation of such multiple assessment formulae would likely be quite expensive. Second, and more fundamental, is the fact that the assessments for all potential projects must be decided on prior to participants' purchase of shares; otherwise, the setting of assessments itself will require a unanimous vote, which seems very costly to implement. On the other hand, a single assessment formula that applies to all prospective projects is unlikely to generate unanimous votes in all cases. If a single formula is nevertheless selected, the closer it comes to generating unanimity, on average, the lower resource expenditures to capture wealth are predicted to be.

We now turn specifically to the development of a condominium.²³ In order to maximize the value of individual units, the condominium developer can choose the assessment method to be employed, the vote allocation scheme and various other features of his project.

The distribution of benefits among units in a condominium is likely to be a function of both the number and characteristics of residents, as well as the heterogeneity in the attributes and value of individual units. For example, a common facility such as a swimming pool may generate benefits nearly proportional to the number of (swimming) residents in a unit. In contrast, the benefits of a central heating system are likely to be proportional to the area of a unit. Maintaining architectural standards and keeping common

22. The total net gain a project generates can be divided in a number of ways, and thus the condition in the text is not sufficient for unanimity. The discussion of the necessary condition suffices here, however.

23. Ellickson [1982] provides a detailed discussion of voting in condominiums and homeowners' associations.

areas attractive may generate benefits roughly in proportion to a unit's value.

An assessment that is uniform across condominium units is easily implemented and will serve well if most future projects in a condominium tend to generate the same benefits to all unit owners. The more uniform the units themselves, the more likely that this condition will be satisfied. It seems plausible that benefits from some future projects will generally be a function of both area and value. If this is the case, the more varied are the units' areas within a condominium, the less the likelihood that uniform assessments will lead to near unanimous votes, and the greater the likelihood that assessments that are proportional to unit size will enhance unanimity. Similarly, the more units vary in value per square foot, the greater the likelihood that the assessment formula will be based on value in order to serve well, and the more we can expect to see assessments by value. Thus, we test below the hypothesis that the greater the variance in value per square foot of units within a condominium, the greater the likelihood of assessments being apportioned by value; the more units vary by size, the greater the probability that assessments will be allocated by area.

The vote allocation selected by the developer will affect both the wealth transfer costs and decision-making costs associated with voting. Holding constant the costs of carrying out the voting process, when benefits are positive and assessments are strictly proportional to benefits, the method of allocating votes is immaterial.²⁴ Since under these conditions it is impossible for some group members to gain at the expense of others, resource allocation within the voting organization is independent of the allocation of votes.²⁵ As we pointed out above, however, deviations from proportionality between assessments and benefits are expected. Decision-making by majority vote allows majorities to form in order to exploit the deviation from proportionality between assessments and benefits, and to gain at the expense of minorities. The ability to form such majorities depends in part on the vote allocation method. Allocating votes by value of investment will not always best prevent majorities from capturing minorities' wealth. For example, if votes were proportional to the value of units but assessments were

24. We abstract from a possible divergence between average and marginal conditions, which, however, does not affect subsequent results.

25. Under the stated conditions, property rights are perfectly defined, and as implied by the Coase Theorem, no resources are spent to capture wealth.

uniform, owners of more valuable units could exploit the voting process to reap gains at the expense of owners of smaller units. The greater the discrepancy between individuals' economic interests (i.e., the assessments and benefits) and the corresponding amount of votes, the greater the opportunity for wealth capture. We predict that votes will tend to be allocated proportionally to the gains or to the assessments. In a stock corporation this condition is clearly satisfied since assessments are zero, and votes are proportional to shares. In the case of a condominium, where gains are costly to gauge, votes are expected to be proportional to assessments.

The vote allocation method will also affect the costs of decision-making within a voting organization, one of which is simply counting votes. In the case of condominiums, uniform allocation of votes allows tallying of votes by a simple show of hands at a meeting. In contrast, votes based on unit values or areas may require tallying of fractional voting shares, as well as producing and policing signed ballots. The greater the number of voting members, the higher these costs will be. Therefore, a bias in favor of uniform allocation of votes is expected. Further, this tendency toward uniform voting is likely to be stronger the greater the number of units in the condominium. Below, in Section IV we test the proposition that deviations in the allocation of votes from the method chosen for assessments are toward uniform voting, and this tendency toward uniform voting is more pronounced the greater the number of units in the condominium.

Another method by which a developer can affect the costs of a voting organization is through control of issues that will be subject to vote. Potentially disputed projects that are undertaken prior to the sale of the assets to individual buyers are simply not subjected to vote. It is therefore expected that developers will undertake ahead-of-time projects whose total net value is positive, but regarding which, opinions are likely to be divided. The clearest example is condominiums with uniform units, where deviations in net benefits among owners should be minimal. In such condominiums it is predicted that amenities are less likely to be installed by the developer. This implication is tested below in Section IV.

One obvious disadvantage of developer-constructed amenities is that they require early commitment, even when waiting may be advantageous. When project benefits are confined to easy-to-identify subsets of buyers, this problem can be avoided by forming "local" governments within the organization. By creating such

subunits with their own assessments and voting allocations, one will avoid the potential capture as well as the difficulty of assuring that the majority of all buyers vote for such projects.²⁶

The final method by which a developer can seek to minimize costly wealth transfers is through the choice of the voting rule. Buchanan and Tullock [1962] argue that the choice of the optimal voting rule depends on the trade-off between "external costs" (mostly individuals' losses from transfers) and decision-making costs. As the voting rule becomes more inclusive (nearer unanimity), external costs decline, since the size of the minority with potential to be exploited by the majority declines. In the extreme case of unanimity, no voter can impose costs on others without their consent, and external costs would be zero.²⁷ There would then be no resources devoted to avoiding wealth transfers. Decision-making costs, however, increase with the percentage of votes required to take action for two reasons. First, the greater the percentage for a given size group, the greater the absolute number of voters who must reach agreement. The larger the absolute size of the requisite coalition, the more costly that coalition is to achieve. And second, strategic bargaining costs due to the "holdout" problem also increase with the inclusiveness of the decision rule. Under unanimity, any single voter can determine the outcome, thereby intensifying the problem of strategic bargaining.

Based on this reasoning, Buchanan and Tullock claim that the smaller the size of the decision-making unit, the more inclusive the voting rule will be. Holding constant the voting rule and the method of assigning votes, the larger the group size, the greater the absolute number of people required for agreement, and the higher the decision-making costs. In addition, Buchanan and Tullock argue that external costs are relatively constant across groups of different sizes. Consequently, the sum of external costs and decision-making costs is an increasing function of group size, and the larger the group, the less inclusive is the optimal voting rule.

Kiesling [1968] challenges the Buchanan-Tullock prediction, arguing that as group size increases, strategic bargaining costs fall,

26. Occasionally, a condominium will adopt a general assessment scheme for most issues, and a special assessment formula for a particular service. For example, at least one Parisian developer adopted a clever assessment formula to cover the cost of running an elevator. In this four-story condominium, ground floor units are not assessed at all; first floor units are assessed half of what second floor units are assessed; and one third that of third floor units.

27. The benefits of a unanimity rule are discussed extensively in Buchanan and Tullock [1962, Ch. 7].

since the absolute number of voters required to block the majority and "hold out" for a more favorable outcome increases with the size of the group. He concludes that overall decision-making costs as a function of group size are therefore ambiguous.

Consistent with our theory, Buchanan and Tullock also argue that the more heterogeneous are the voters in a group, the more inclusive the voting rule will be. They reason that while decision-making costs will be high for heterogeneous groups, the potential for exploitation of the minority by the majority will also be higher the more divergent the interests of group members, leading to the choice of a more inclusive voting rule.

To summarize, developers of voting organizations will select the allocation of assessments and votes to members of the organization in a way that minimizes the sum of the costs associated with wealth transfer and decision-making. Issues for which these costs are "too high" will be excluded by the developer from the domain of voting. Further, for those issues that are subject to vote, the choice of the voting rule will depend on the heterogeneity of interests within the voting group.

IV. EMPIRICAL TESTS: THE CONSTITUTIONS OF CONDOMINIUM OWNER ASSOCIATIONS

The "declaration" is the constitution of a condominium homeowner's association. The declaration is written by the developer of the condominium and must be filed with the clerk of the county in which the condominium is located. Subject to certain restrictions in state condominium acts,²⁸ the developer is free to choose the declaration's provisions. Thus, the declaration should reflect his attempts to lower the costs of the voting organization in order to maximize the value of the condominium units he is selling. By law, the declaration must describe the physical features of the individual units and common amenities, and specify the allocation of votes and assessments among condominium unit owners. Consequently, many of the implications of our theory of constitutional choice can be tested by analyzing the declarations of condominiums.

Our data include declarations from condominiums in the counties of Bernalillo, Santa Fe, and Lincoln in New Mexico. Albuquerque, the largest city in New Mexico, is located in Bernalillo.

28. For a review of state condominium statutes, see Ellickson [1982] and references therein.

lillo County. The third largest city, Santa Fe, is located in Santa Fe County, where due to high land values, the number of condominiums is greater than in any other county in the state. The resort town of Ruidoso is in Lincoln County where there are also a large number of condominiums. The total sample from these three counties includes the declarations of 120 condominiums. Of these, 101 are residential, seven are offices, four are commercial, five are mixed use, and the remaining three could not be identified.²⁹ Information contained in this sample of declarations is used to test predictions made in the previous sections.

Consider first the effect of the diversity in attribute levels on the choice of the assessment formula among individual units within a condominium. The theory predicts that the assessment allocation chosen will be the one most nearly proportional to the benefits received. Virtually all condominiums in the sample assign assessments according to one of three formulas: (1) uniform assessments across units, (2) assessments based on the relative area of units, or (3) assessments based on the relative value of units.³⁰ If benefits are related to both area and value, then our model predicts that the greater the variance in unit areas or unit values, the less likely it is that uniform assessments will be assigned. Similarly, the greater the variance in value per square foot, the more likely assessments are to be based on unit values rather than on unit areas.

In order to test these hypotheses, a multiple logit regression was performed to estimate the effects of differences in the variabil-

29. For Bernalillo County, the sample represents all 40 condominium declarations filed since the initial New Mexico condominium legislation of 1963 through November 1984. For Santa Fe County, records were searched for the period April 1974 through April 1984. This period encompasses the vast majority of declaration filings. A total of 108 declarations were filed during the period. However, for those declarations filed in July 1980 or later, condominiums lacking area or value data are not included in the sample. This excluded 42 condominiums, leaving a sample of 66. The average size of condominiums in Santa Fe County was 9.2 units per condominium compared with 75.2 for Bernalillo County. In Lincoln County only those declarations containing both area and value data were included in the sample. Thus, the sample from Lincoln County includes only 14 out of the 37 condominium declarations which had been filed through August 1985. The average size of condominiums in Lincoln County was 22.1 units. Total sample size is 120 (40 + 66 + 16).

30. Condominium developers were legally constrained to adopt one of these three allocation formulas from 1975 through 1981. During that period New Mexico law required that voting "be based upon: the square or cubic footage in the unit as a percentage of the square or cubic footage in all of the units; the value, as that term is defined in the declaration, of the unit as a percentage of the value of all of the units; or the unit itself as a percentage of all of the units in the property" [New Mexico Statutes Annotated 1978, § 47-7-12]. Before and after this period any allocation scheme could be used.

TABLE I
ASSESSMENT ALLOCATION REGRESSION

Dependent variable	Constant	Standard deviation in area of units	Standard deviation in value per square foot of units
Unweighted sample			
1. $\log (P_A/P_U)$	-2.934*** (-3.86)	0.004 (1.12)	2.023*** (2.65)
2. $\log (P_V/P_U)$	-1.867** (-2.88)	0.001 (0.37)	2.066*** (2.72)
3. $\log (P_A/P_V)$	-1.067* (-1.81)	0.003 (1.62)	-0.043 (-0.28)
Weighted sample			
1. $\log (P_A/P_U)$	-2.045** (-2.06)	0.005 (0.92)	1.711*** (3.04)
2. $\log (P_V/P_U)$	-2.023* (-1.71)	0.002 (0.39)	1.753*** (2.86)
3. $\log (P_A/P_V)$	-0.022 (-0.04)	0.003 (1.15)	-0.041 (-0.24)
Total number of observations = 52			
Chi-squared statistic = $-2 \times (L_0 - L_{\max}) = 34.04^{***}$ (unweighted sample)			
= 31.39*** (weighted sample)			

Note. U = uniform, A = area-based, and V = value-based assessments. Asymptotic t -ratios are in parentheses.

*Significant at the 0.10 level; **significant at the 0.05 level; ***significant at 0.01 level in a two-tailed test.

ity of unit areas and unit values among condominiums on the relative probabilities of selecting uniform, area-based, and value-based assessment schemes. The original sample, consisting of 52 condominiums for which data on both unit values and unit areas were available, was weighted to account for sample selection bias resulting from an oversampling of condominiums with value-based assessment schemes.³¹ Results for both the original and the weighted sample are reported in Table I.

The dependent variables in the three equations of the multiple logit model are $\log (P_A/P_U)$, the log of the probability of selecting

31. As one might expect, condominiums with uniform or area-based assessments were much less likely to report unit values in their declarations than were condominiums with value-based assessment schemes. Therefore, condominiums with value-based assessments were overrepresented in the original sample and condominiums with area-based or uniform assessment formulas were underrepresented. To correct for this, the sample was weighted so that the proportions of condominiums with uniform, area-based, and value-based assessments were the same as in the population. Details of the weighting procedure are specified in Greene [1985, pp. 148-152], and Manski and McFadden [1981].

area-based assessments relative to the probability of selecting uniform assessments; $\log (P_V/P_U)$, the log of the probability of choosing value-based assessments relative to the probability of selecting uniform assessments; and $\log (P_A/P_V)$, the log of the probability of selecting an area-based assessment formula relative to the probability of choosing an assessment scheme based on unit values. The coefficients of the independent variables in the third equation are not independently estimated. Rather, since $\log (P_A/P_V) = \log (P_A/P_U) - \log (P_V/P_U)$, they are simply the difference between the coefficients in the first and second equations.³²

The coefficient on the area variability variable is positive in the first and third equations, as expected, indicating that increases in the variance in unit areas tend to increase the probability that an area-based assessment scheme will be chosen relative to the likelihood that uniform assessments or value-based assessments will be selected. However, the area variability coefficient is insignificant in the first equation and only significant at about the 10 percent level in the unweighted third equation.³³

The coefficient on the value variance variable is positive and highly significant in the second equation, as expected, indicating the probability of choosing value-based assessments relative to the probability of selecting uniform assessments increases with increases in the standard deviation in unit values. The value variance coefficient is negative in the third equation, as predicted, indicating that increases in the variability of unit values within a condominium tend to decrease the likelihood that area-based assessments will be chosen relative to the probability that value-based assessments are selected. This effect is not significantly different from zero, however.

The negative and significant coefficients on the constant term in the first and second equations indicate that holding constant the variance in unit attributes, uniform assessments tend to be preferred relative to area-based or value-based assessments.

Our theory also predicts that votes will be assigned in order to conform to owners' gains and costs from new projects. In particular,

32. For a discussion of the precise formulation of the multiple logit model, see Pindyck and Rubinfeld [1981].

33. The lack of significance of the coefficient on the area variability variable may be due to collinearity between the area and value variability measures. A simple binary logit regression of $\log (P_A/P_U)$ on the standard deviation in unit areas and a constant yielded a positive and significant coefficient on the area variability measure ($t = 2.45$).

the method chosen to allocate votes within condominiums is expected to be the same as that chosen for assessments. Other than the potential for dissipation, however, the choice of voting assignments is also influenced by the cost of utilizing the voting mechanism. It is expected that, *ceteris paribus*, a one-vote-per-unit rule would be preferred, since counting hands is easier than adding up fractional votes. Further, the reduction in decision-making costs resulting from use of uniform voting would be greater with a greater number of voters. Therefore, the larger the size of the voting group, the more likely uniform voting should be.

These predictions are supported by the data. Table II shows that the vast majority of condominiums in the sample employ the same rule for assessments and voting. Further, *all* of the deviations between the voting method and the method of assessment are toward one vote per unit. Table III reports the results of a binary logit regression where the dependent variable equals one if the assignment of votes was uniform and the assignment of assessments nonuniform, zero otherwise. As predicted, the probability of choosing a uniform voting scheme, despite nonuniform assessments, increased with the number of units in the condominium.

The developer may also choose which features to provide in advance, thereby removing them from the domain of voting decisions. Our theory predicts that the greater the potential voting costs associated with decisions about a project, the more likely it is that the project will be undertaken ahead of time. The potential wealth transfer costs of voting will be greater the greater the variation in net benefits among members. Assuming that benefits are related to

TABLE II
ASSIGNMENT OF VOTING RIGHTS BY METHOD OF ASSESSMENT

Method of assessment	Assignment of voting rights			
	By unit	By area	By value	Total
By unit	35 (100.0)	0 (0.0)	0 (0.0)	35 (100.0)
By area	9 (18.0)	41 (82.0)	0 (0.0)	50 (100.0)
By value	1 (3.7)	0 (0.0)	26 (96.3)	27 (100.0)
All methods	45 (40.2)	41 (36.6)	26 (23.2)	112 (100.0)

Note. Row percentages are in parentheses.

TABLE III
UNIFORM VOTING REGRESSION

Independent variable	Predicted sign	Coefficient (t-ratio)
Constant		-2.867*** (-6.49)
Number of units	+	0.011*** (2.66)
Number of observations = 111		
Chi-squared statistic = $2 \times (L_0 - L_{\max}) = 6.69^{***}$		

Note. ***Significant at the 0.01 level in a two-tailed test.

the physical features of units, then in condominiums with units of equal size and value, variation in net benefits will be nearly absent, and we therefore expect that amenities are less likely to be installed by the developer.

The number of developer-installed common amenities will also depend on the demand for such features. To the extent that amenities like swimming pools and lounges are quasi-public goods, net benefits per unit and total demand should increase with the number of units in the condominium. In addition, recreational amenities may be more highly valued in resort areas than in year-round residences. The demand for amenities may also be related to the incomes of unit owners. As incomes rise, it seems that demand for features like hot tubs or lounge areas is likely to increase. However, higher income owners may prefer to consume some of these features individually rather than in common (e.g., a bathtub with a jacuzzi rather than a common hot tub or a large living room rather than a lounge). Therefore, while it is expected that the number of developer-installed amenities is likely to be higher in larger condominiums and resort condominiums, the effect of income on the number of amenities is uncertain.

To test these predictions, we performed a least squares regression in which the dependent variable equaled the total number of amenities (swimming pools, hot tubs, tennis courts, recreation rooms, and meeting rooms) erected by the developer.³⁴ Only residential condominiums were included in the regression since our theory does not suggest difficulties in adding amenities to nonresi-

34. These five categories of amenities are the only common amenities that exist in any of the condominiums in the sample.

TABLE IV
AMENITIES REGRESSION
(RESIDENTIAL CONDOMINIUMS ONLY)

Independent variable	Predicted sign	Coefficient (t-ratio)
Constant		1.069 (0.45)
Condominiums with uniform units	—	—0.413** (—2.05)
Log of number of units	+	0.353*** (3.82)
Log of average unit value	?	—0.150 (—0.66)
Resort condominiums	+	0.053 (0.24)
Number of observations = 47		$R^2 = 0.36$

Note. **Significant at the 0.05 level; *** significant at the 0.01 level in a two-tailed test.

dential condominiums. The first independent variable takes on a value of one if units in the condominium have equal areas and values, zero otherwise. The second and third explanatory variables are the natural logs of the number of units and the average real value of units, respectively. The latter is a proxy for the income of condominium owners. The last independent variable is a dummy variable indicating condominiums located in the resort area of Ruidoso. In the amenities regression, summarized in Table IV, the coefficients of all of the variables have the expected signs. The coefficients on both the uniformity and size variables are highly significant.³⁵

Finally, supermajority rules are asserted to prevent the majority from exploiting the minority. There are several distinct forces that may influence such voting rules. The first relates to capital improvements such as building a swimming pool or remodeling the exterior of buildings. As discussed above, if there is great potential for conflict, these projects may be undertaken in advance by the developer. Once the facilities are in place, the probability of

35. The variable measuring average unit values could be correlated with the error term since unit values are in part determined by the number of common amenities. To account for this possibility, a two-stage least squares regression with unit area in the instrument list was run. The results were virtually identical to the ordinary least squares estimates reported in Table IV.

additional improvements would be reduced.³⁶ Consequently, it is expected that the voting rule for improvements would be less inclusive for condominiums with facilities already in existence. In addition, in small developments the per unit cost of major improvements is high, making the initiation of such issues unlikely. If this were the only effect of the number of units, we would expect the inclusiveness of the voting rule to increase with the number of units in the condominium. However, Buchanan and Tullock predict that due to the higher decision-making costs encountered by large groups, the inclusiveness of the voting rule will be inversely related to group size. Therefore, the net effect of the number of units on the optimal voting rule is indeterminate. Buchanan and Tullock also predict that the more homogeneous the voting group, the less restrictive the voting rule. Last, our theory predicts that commercial and office condominiums are more apt to have less restrictive rules than those that exist in residential condominiums, since benefits in the former are primarily pecuniary.

Table V shows the results of a least squares regression where the dependent variable is the minimum percentage vote required to pass capital improvement expenditures voted on by unit owners.³⁷ Condominiums with less than ten units were excluded, since differences in voting percentages have little meaning for very small groups. The coefficient on the first independent variable, the number of units within the condominium, is positive and significantly different from zero. This result indicates that the effect of increases in the number of units on the likelihood of capital improvement issues dominates the effect such increases in the number of voters has on decision-making costs. The second independent variable is a dummy variable with a value of one if the condominium is located in the resort area of Ruidoso, and zero if located elsewhere. Decision-making costs tend to be high in condominiums located in such resort areas, since many owners are not full-time residents. Further, it is common for resort condominium owners to rent out their units when they are not in residence,

36. A referee points out that the initial placement of common amenities by a developer could be an indicator of unit owners' having a relatively high demand for these goods. If this is the case, then the probability of future debate over installation of amenities could be positively correlated with the existence of developer-installed amenities.

37. In some condominiums the board of directors can make capital improvement decisions independently of unit owners, though such decisions can be challenged by unit owners. The voting rules for representative bodies are beyond the scope of this paper. However, see Sass [1988].

TABLE V
VOTING RULE REGRESSION
(CONDOMINIUMS WITH TEN OR MORE UNITS)

Independent variable	Predicted sign	Coefficient (t-ratio)
Constant		50.529*** (10.5)
Log of number of units	?	2.433* (1.84)
Resort condominium	—	—5.617* (—1.87)
Nonresidential condominium	—	—6.160 (—1.33)
Mixed use condominium	+	7.764 (1.37)
Number of swimming pools and hot tubs	—	—6.692** (—2.29)
Number of recreation or activities buildings	—	7.792** (2.04)
Number of meeting rooms and lounges	—	—4.612 (—1.15)
Number of tennis facilities	—	—8.178 (—1.40)
Number of observations = 60		$R^2 = 0.28$

Note. *Significant at the 0.10 level; **significant at the 0.5 level; ***significant at the 0.01 level in a two-tailed test.

thereby creating pecuniary as well as nonpecuniary benefits. For both of these reasons, it is expected that resorts in Ruidoso would tend to have less inclusive voting rules; indeed the coefficient on resort condominiums is negative and significant. Since commercial and office condominiums provide mainly pecuniary benefits to their unit owners, they are expected to have less inclusive voting rules than do residential condominiums which provide substantial nonpecuniary benefits. The dummy variable indicating commercial and office condominiums, is negative, as expected. However, this effect was not significant. The fourth variable is a dummy indicating mixed use condominiums that contain both residential and commercial or office units. Due to the heterogeneity of ownership, such condominiums are expected to have more restrictive voting rules. The coefficient on the mixed use variable is positive, but insignificant. The next four independent variables measure the influence of various amenities on the optimal voting rule. We predict that construction of additional amenities is less likely to be a voting issue

if some amenities are already in place, thereby making possible the use of a less inclusive voting rule. The coefficients of all four variables are expected to be negative. The results of this test are inconclusive. The coefficient for the number of recreation or activities buildings installed is positive.³⁸ Of the other three, only the coefficient on the number of swimming pools and hot tubs is significantly different from zero.

V. SUMMARY

The use of voting to determine the allocation of resources is an alternative to use of the price mechanism. Market exchange separates ownership from use, whereas voting organizations are often owned by the same people who consume the services produced by the organization.

The study of allocation by political voting organizations is problematic because the interests of political voters are difficult to untangle. However, it is relatively easy to analyze the behavior of developers who set up the physical operations and structure the constitutions of private voting groups, and then sell voting shares in these organizations. In particular, the more appealing the constitutions of such developers' organizations to prospective buyers, the higher will be developers' gains, since the net revenues elicited from buyers are then also higher.

When resources are allocated by voting, costs associated with majorities' opportunities for capturing wealth (and minorities' preventative efforts) and difficulties of making decisions arise. Developers will attempt to convince potential buyers that these costs are likely to be low.

It is argued that developers will gain (1) by closely matching projects' assessments to projects' benefits; (2) by closely matching projects' net benefits to the allocation of votes among members; (3) by providing ahead of time (or excluding from the organization's jurisdiction) projects that, if proposed later on, are likely to be divisive; and (4) by requiring more inclusive voting rules (larger

38. The positive and significant coefficient on the recreation building variable may be due to the potential capital improvement expenditures of equipping such structures. While swimming pools and tennis courts are essentially complete once installed, potentially divisive decisions regarding purchases of recreational equipment must be made if a recreation building is erected.

majorities) for future projects whose total net value is large, but that make easy targets for the capture of wealth.

The predictions of the theory are tested for one type of voting organization: condominium owners' associations. The decision to study condominiums is the result of the availability of data on their physical features and constitutions and of the wide range of voting and assessment rules they employ. Consistent with the theory, it is first found that the greater the variance in value per square foot among condominium units, the more likely will be the selection of area-based assessments. Likewise, the probability of selecting value-based assessments tends to increase with the variance in unit value per square foot. Second, votes tend to be allocated in the same manner as assessments; when they do deviate, it is always toward the lower cost system of uniform voting. The probability of selecting uniform voting, despite nonuniform assessments, increases with the number of units in the condominium. Third, the greater the potential for divisiveness of a project, the higher the likelihood of developers undertaking them in advance. The construction of common amenities in a condominium is an example of such a project. We found that when condominium units are uniform and thus the divergence in owner's net benefits are low, fewer amenities were put in place by the developer. Finally, we found the inclusiveness of the voting rule selected tends to increase with increases in the heterogeneity of the voting group and the degree to which benefits are nonpecuniary. While not all of these results are significant, all but one of the effects we were able to test for are in the predicted directions, and taken as a whole, they strongly support our theory. Thus, our embryonic theory of constitution, based on maximizing projects' values to their owners, and derived from profit maximization by project developers, gains considerable support from the empirical results.

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